

1. Create Database and Books Table

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' and 'INSTANCE' sections. The main editor displays a SQL script with the following queries:

```
34
35 INSERT INTO books(title, isbn, published_year)
36 VALUES ('Fake Copy', '9780061122415', 2000);
37
38 #practical-2
39 CREATE DATABASE IF NOT EXISTS book_db;
40 USE book_db;
41
42 CREATE TABLE IF NOT EXISTS Books (
43     book_id INT PRIMARY KEY,
44     title VARCHAR(100) NOT NULL,
45     isbn VARCHAR(20) UNIQUE,
46     published_year INT CHECK (published_year < 2027)
47 );
```

The 'Output' tab at the bottom shows the execution results:

#	Time	Action	Message
6	22:53:08	DESCRIBE members	3 row(s) returned
7	22:54:24	INSERT INTO bookstitle, isbn, published_year VALUES (The Alchemist, '978006...	3 row(s) affected Records: 3 Duplic
8	22:56:33	INSERT INTO membersfull_name, email VALUES (Anil Kumar, 'anil.kumar@exa...	3 row(s) affected Records: 3 Duplic
9	22:57:36	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
10	22:58:07	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO bookstitle, isbn, published_year VALUES (Fake Copy, '978006112...	Error Code: 1062, Duplicate entry '9
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected

This screenshot shows the same MySQL Workbench interface, but the SQL script is now highlighted in blue, indicating it has been executed. The 'Output' tab shows the results of the execution:

#	Time	Action	Message
7	22:54:24	INSERT INTO bookstitle, isbn, published_year VALUES (The Alchemist, '978006...	3 row(s) affected Record
8	22:56:33	INSERT INTO membersfull_name, email VALUES (Anil Kumar, 'anil.kumar@exa...	3 row(s) affected Record
9	22:57:36	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
10	22:58:07	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO bookstitle, isbn, published_year VALUES (Fake Copy, '978006112...	Error Code: 1062, Duplic
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected
14	23:13:30	USE book_db	0 row(s) affected

This screenshot shows the MySQL Workbench interface with the SQL script highlighted in blue. The 'Output' tab shows the results of the execution:

#	Time	Action	Message
8	22:56:33	INSERT INTO membersfull_name, email VALUES (Anil Kumar, 'anil.kumar@exa...	3 row(s) affected Record
9	22:57:36	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
10	22:58:07	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO bookstitle, isbn, published_year VALUES (Fake Copy, '978006112...	Error Code: 1062, Duplic
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected
14	23:13:30	USE book_db	0 row(s) affected
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected

2. Insert Books

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main editor displays the following SQL queries:

```
40 USE book_db;
41
42 CREATE TABLE IF NOT EXISTS Books (
43     book_id INT PRIMARY KEY,
44     title VARCHAR(100) NOT NULL,
45     isbn VARCHAR(20) UNIQUE,
46     published_year INT CHECK (published_year < 2027)
47 );
48
49 INSERT INTO Books (book_id, title, isbn, published_year)
50 VALUES
51 (1, 'The Great Gatsby', '9780743273565', 1925),
52 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
53 (3, 'In The Cave', '9780451524935', 1949);
```

The 'Output' tab shows the results of the queries:

#	Time	Action	Message
9	22:57:36	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
10	22:58:07	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO booktitle (isbn, published_year) VALUES (Fake Copy, '9780061120084')	Error Code: 1062 Duplicat...
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected
14	23:13:30	USE book_db	0 row(s) affected
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Reco...

3. Display Books

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main editor displays the following SQL queries:

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49 INSERT INTO Books (book_id, title, isbn, published_year)
50 VALUES
51 (1, 'The Great Gatsby', '9780743273565', 1925),
52 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
53 (3, 'In The Cave', '9780451524935', 1949);
54
55 SELECT * FROM Books;
```

The 'Result Grid' shows the results of the query:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	In The Cave	9780451524935	1949

The 'Output' tab shows the results of the queries:

#	Time	Action	Message
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO booktitle (isbn, published_year) VALUES (Fake Copy, '9780061120084')	Error Code: 1062 Duplicat...
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected
14	23:13:30	USE book_db	0 row(s) affected
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Reco...
17	23:16:56	SELECT * FROM Books	3 row(s) returned
18	23:18:22	CREATE TABLE IF NOT EXISTS Members (member_id INT PRIMARY KEY, ...	0 row(s) affected

4. Create Members Table

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main editor displays the following SQL queries:

```
40 INSERT INTO Books (book_id, title, isbn, published_year)
41 VALUES
42 (1, 'The Great Gatsby', '9780743273565', 1925),
43 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
44 (3, 'In The Cave', '9780451524935', 1949);
45
46 SELECT * FROM Books;
47
48 CREATE TABLE IF NOT EXISTS Members (
49     member_id INT PRIMARY KEY,
50     full_name VARCHAR(100),
51     email VARCHAR(100) UNIQUE
52 );
```

The 'Output' tab shows the results of the queries:

#	Time	Action	Message
11	22:59:12	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
12	23:00:11	INSERT INTO booktitle (isbn, published_year) VALUES (Fake Copy, '9780061120084')	Error Code: 1062 Duplicat...
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) affected
14	23:13:30	USE book_db	0 row(s) affected
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Reco...
17	23:16:56	SELECT * FROM Books	3 row(s) returned
18	23:18:22	CREATE TABLE IF NOT EXISTS Members (member_id INT PRIMARY KEY, ...	0 row(s) affected

5. Insert Members

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main query editor displays the following SQL code:

```
54
55 SELECT * FROM Books;
56
57 CREATE TABLE IF NOT EXISTS Members (
58     member_id INT PRIMARY KEY,
59     full_name VARCHAR(100),
60     email VARCHAR(100) UNIQUE
61 );
62
63 INSERT INTO Members (member_id, full_name, email)
64 VALUES
65 (101, 'Dikshita', 'dikshita@gmail.com'),
66 (102, 'Emma Wilson', 'emma.wilson@email.com'),
67 (103, 'Michael Brown', 'michael.brown@email.com');
```

The 'Output' tab shows the execution results:

#	Time	Action	Message
12	23:00:11	INSERT INTO books(title, isbn, published_year) VALUES ('Fake Copy', '978006112...	Error Code:
13	23:12:15	CREATE DATABASE IF NOT EXISTS book_db	1 row(s) aff
14	23:13:30	USE book_db	0 row(s) aff
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) aff
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) aff
17	23:16:56	SELECT * FROM Books LIMIT 0, 1000	3 row(s) ret
18	23:18:22	CREATE TABLE IF NOT EXISTS Members (member_id INT PRIMARY KEY, ...	0 row(s) aff
19	23:19:18	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Dikshita', 'd...	3 row(s) aff

6. Display Members

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main query editor displays the following SQL code:

```
63 INSERT INTO Members (member_id, full_name, email)
64 VALUES
65 (101, 'Dikshita', 'dikshita@gmail.com'),
66 (102, 'Emma Wilson', 'emma.wilson@email.com'),
67 (103, 'Michael Brown', 'michael.brown@email.com');
68
69 SELECT * FROM Members;
```

The 'Result Grid' shows the following data:

member_id	full_name	email
101	Dikshita	dikshita@gmail.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

The 'Members' table is shown with 7 rows.

7. Create Loans Table

The screenshot shows the MySQL Workbench interface. The left sidebar contains the 'MANAGEMENT' section with options like Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, and Data Import/Restore. The 'INSTANCE' section includes Startup / Shutdown, Server Logs, and Options File. The 'PERFORMANCE' section includes Dashboard, Performance Reports, and Performance Schema Setup. The 'Administration' section includes Schemas and Information. The 'Information' section shows 'No object selected'.

The main query editor displays the following SQL code:

```
64 VALUES
65 (101, 'Dikshita', 'dikshita@gmail.com'),
66 (102, 'Emma Wilson', 'emma.wilson@email.com'),
67 (103, 'Michael Brown', 'michael.brown@email.com');
68
69 CREATE TABLE IF NOT EXISTS Loans (
70     loan_id INT PRIMARY KEY,
71     member_id INT,
72     book_id INT,
73     loan_date DATE,
74     FOREIGN KEY (member_id) REFERENCES Members(member_id),
75     FOREIGN KEY (book_id) REFERENCES Books(book_id)
76 );
77
```

The 'Output' tab shows the execution results:

#	Time	Action	Message
14	23:13:30	USE book_db	0 row(s) affected
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records: 3 Duplic
17	23:16:56	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
18	23:18:22	CREATE TABLE IF NOT EXISTS Members (member_id INT PRIMARY KEY, ...	0 row(s) affected
19	23:19:18	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Dikshita', 'd...	3 row(s) affected Records: 3 Duplic
20	23:20:11	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
21	23:21:17	CREATE TABLE IF NOT EXISTS Loans (loan_id INT PRIMARY KEY, memb...	0 row(s) affected

8. Insert Loans

The screenshot shows the MySQL Workbench interface with a query editor containing an `INSERT INTO` statement for the `Loans` table. The query is as follows:

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date)
VALUES
(1, 101, 1, '2025-01-05'),
(2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'),
(4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'),
(6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'),
(8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'),
(10, 101, 1, '2025-04-01');
```

The Output tab shows the execution results, indicating that 10 rows were affected by the insert operation.

#	Time	Action	Message
15	23:14:59	CREATE TABLE IF NOT EXISTS Books (book_id INT PRIMARY KEY, title V...	0 row(s) affected
16	23:16:10	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records
17	23:16:56	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
18	23:18:22	CREATE TABLE IF NOT EXISTS Members (member_id INT PRIMARY KEY, ...	0 row(s) affected
19	23:19:18	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Dikshita', 'di...	3 row(s) affected Records
20	23:20:11	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
21	23:21:17	CREATE TABLE IF NOT EXISTS Loans (loan_id INT PRIMARY KEY, memb...	0 row(s) affected
22	23:22:19	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1,...	10 row(s) affected Record

9. Display Loans

The screenshot shows the MySQL Workbench interface with a query editor containing the following query:

```
SELECT * FROM Loans;
```

The Output tab displays the results in a Result Grid, showing 10 rows of loan data.

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10. INNER JOIN

MySQL Workbench interface showing a query window with the following SQL code:

```

95 m.full_name AS Member_Name,
96 b.title AS Book_Title
97 FROM Loans l
98 INNER JOIN Members m
99 ON l.member_id = m.member_id
100 INNER JOIN Books b
101 ON l.book_id = b.book_id

```

The Result Grid shows the following data:

Member_Name	Book_Title
Dikshita	The Great Gatsby
Dikshita	To Kill a Mockingbird
Dikshita	In The Cave
Dikshita	The Great Gatsby
Emma Wilson	To Kill a Mockingbird

11. GROUP BY and ORDER BY

MySQL Workbench interface showing a query window with the following SQL code:

```

102
103 SELECT
104 published_year,
105 COUNT(book_id) AS Total_Books
106 FROM Books
107 GROUP BY published_year
108 ORDER BY published_year;

```

The Result Grid shows the following data:

published_year	Total_Books
1925	1
1949	1
1960	1

12. Create Donation History

MySQL Workbench interface showing the creation of a table and the execution of several queries. The SQL code in the query window is:

```

97 FROM Loans l
98 INNER JOIN Members m
99 ON l.member_id = m.member_id
100 INNER JOIN Books b
101 ON l.book_id = b.book_id
102
103 CREATE TABLE IF NOT EXISTS Donation_History (
104 donation_id INT PRIMARY KEY,
105 book_id INT,
106 donor_name VARCHAR(100),
107 donation_date DATE,
108 FOREIGN KEY (book_id) REFERENCES Books(book_id)
109 )
110

```

The Output window shows the following messages:

```

20 23:20:11 SELECT * FROM Members LIMIT 0, 1000 3 rows returned
21 23:21:17 CREATE TABLE IF NOT EXISTS Loans ( loan_id INT PRIMARY KEY, memb... 0 rows affected
22 23:22:19 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1... 10 rows affected Records: 10 Duplicates: 0 Warnings: 0
23 23:23:07 SELECT * FROM Loans LIMIT 0, 1000 10 rows returned
24 23:24:01 SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l... Error Code: 1064 'You have an error in your SQL syntax; check th... 10 rows returned
25 23:24:07 SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l... 10 rows returned
26 23:24:52 SELECT published_year, COUNT(book_id) AS Total_Books FROM Books G... 3 rows returned
27 23:25:32 CREATE TABLE IF NOT EXISTS Donation_History ( donation_id INT PRIMARY... 0 rows affected

```

13. Transaction

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
109 FOREIGN KEY (book_id) REFERENCES Books(book_id)
110 );
111
112 START TRANSACTION;
113
114 INSERT INTO Books (book_id, title, isbn, published_year)
115 VALUES (4, 'Animal Farm', '9780451526342', 1945);
116
117 INSERT INTO Donation_History
118 (donation_id, book_id, donor_name, donation_date)
119 VALUES
120 (1, 4, 'Raj Kumar', CURDATE());
121
122 COMMIT;
```

The Output tab shows the execution results:

#	Time	Action	Message
24	23:24:01	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans L...	Error Code: 1064. You have an error in your SQL syntax; check the...
25	23:24:07	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans L...	10 row(s) returned
26	23:24:52	SELECT published_year, COUNT(Book_id) AS Total_Books FROM Books G...	3 row(s) returned
27	23:25:32	CREATE TABLE IF NOT EXISTS Donation_History (donation_id INT PRIMARY...	0 row(s) affected
28	23:26:37	START TRANSACTION	0 row(s) affected
29	23:26:37	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Far...	1 row(s) affected
30	23:26:37	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected
31	23:26:37	COMMIT	0 row(s) affected

14. Create Index

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
112 START TRANSACTION;
113
114 INSERT INTO Books (book_id, title, isbn, published_year)
115 VALUES (4, 'Animal Farm', '9780451526342', 1945);
116
117 INSERT INTO Donation_History
118 (donation_id, book_id, donor_name, donation_date)
119 VALUES
120 (1, 4, 'Raj Kumar', CURDATE());
121
122 COMMIT;
123
124 CREATE INDEX idx_books_isbn
125 ON Books(isbn);
```

The Output tab shows the execution results:

#	Time	Action	Message
25	23:24:07	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans L...	10 row(s) returned
26	23:24:52	SELECT published_year, COUNT(Book_id) AS Total_Books FROM Books G...	3 row(s) returned
27	23:25:32	CREATE TABLE IF NOT EXISTS Donation_History (donation_id INT PRIMARY...	0 row(s) affected
28	23:26:37	START TRANSACTION	0 row(s) affected
29	23:26:37	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Far...	1 row(s) affected
30	23:26:37	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected
31	23:26:37	COMMIT	0 row(s) affected
32	23:27:22	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Record

15. Search Using ISBN

The screenshot shows the MySQL Workbench interface with a query window containing the following SQL code:

```
121
122 COMMIT;
123
124 CREATE INDEX idx_books_isbn
125 ON Books(isbn);
126
127 SELECT *
```

The Output tab shows the execution results in a table format:

book_id	title	isbn	published_year
3	In The Cave	9780451524935	1949
4	Animal Farm	9780451526342	1945

Below the table, it indicates "Books 11 x" and "Output" with "Action Output" selected.

```
CREATE DATABASE IF NOT EXISTS book_db;
```

```
USE book_db;
```

```
CREATE TABLE IF NOT EXISTS Books (
```

```
    book_id INT PRIMARY KEY,
```

```
    title VARCHAR(100) NOT NULL,
```

```
    isbn VARCHAR(20) UNIQUE,
```

```
    published_year INT CHECK (published_year < 2027)
```

```
);
```

```
CREATE TABLE IF NOT EXISTS Members (
```

```
    member_id INT PRIMARY KEY,
```

```
    full_name VARCHAR(100),
```

```
    email VARCHAR(100) UNIQUE
```

```
);
```

```
CREATE TABLE IF NOT EXISTS Loans (
```

```
    loan_id INT PRIMARY KEY,
```

```
    member_id INT,
```

```
    book_id INT,
```

```
    loan_date DATE,
```

```
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
```

```
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

```
);
```

```
CREATE TABLE IF NOT EXISTS Donation_History (
```

```
    donation_id INT PRIMARY KEY,
```

```
    book_id INT,
```

```
    donor_name VARCHAR(100),
```

```
    donation_date DATE,
```

```
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

);

INSERT INTO Books VALUES

(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, 'In The Cave', '9780451524935', 1949);

INSERT INTO Members VALUES

(101, 'Dikshita', 'dikshita@gmail.com'),
(102, 'Emma Wilson', 'emma.wilson@email.com'),
(103, 'Michael Brown', 'michael.brown@email.com');

INSERT INTO Loans VALUES

(1, 101, 1, '2025-01-05'),
(2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'),
(4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'),
(6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'),
(8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'),
(10, 101, 1, '2025-04-01');

INSERT INTO Books VALUES

(4, 'Animal Farm', '9780451526342', 1945);

INSERT INTO Donation_History

VALUES (1, 4, 'Raj Kumar', CURDATE());

CREATE INDEX idx_books_isbn

```
ON Books(isbn);
```

```
SELECT * FROM Books;
```

```
SELECT * FROM Members;
```

```
SELECT * FROM Loans;
```

```
SELECT
```

```
    m.full_name AS Member_Name,
```

```
    b.title AS Book_Title
```

```
FROM Loans l
```

```
INNER JOIN Members m ON l.member_id = m.member_id
```

```
INNER JOIN Books b ON l.book_id = b.book_id;
```

```
SELECT
```

```
    published_year,
```

```
    COUNT(book_id) AS Total_Books
```

```
FROM Books
```

```
GROUP BY published_year
```

```
ORDER BY published_year;
```

```
SELECT *
```

```
FROM Books
```

```
WHERE isbn = '9780451524935';
```